6.12.2 Exhaust Pipe

Exhaust pipe is the one which connects exhaust manifold and muffler. It is around 5cm in radius and 1cm in thickness generally. The exhaust pipe arrests the vibrations to the muffler which is coming from the engine.

6.12.3 Exhaust Mufflers

The exhaust mufflers are used to send heat, gas and sound without any destruction. To do these the mufflers have to perform the following operations.

1. Reduce the temperature of exhaust gas
2. Reduce the speed of the exhaust gas
3. Reduce the sound of the exhaust gas
4. Reduce the strength hot and unburned gas

Different types of mufflers are used to perform the above functions. They are of

a) Baffle type
b) Wave cancellation type
c) Resonance type
d) Absorber type
e) Combined absorber and resonance type

6.12.3(a) Baffle type

This type of mufflers will be in cylindrical shape. This will be divided into many rooms. Hence the exhaust gas speed gets reduced and the energy also gets reduced. Fig 6.12.3(a) shows the baffle type muffler.

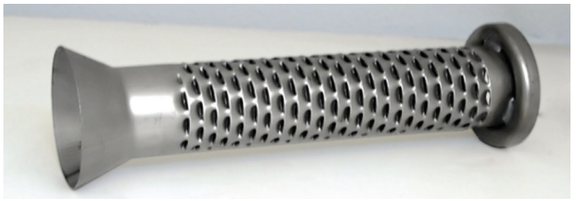


Figure 6.12.3(a) Baffle Type Muffler

6.12.3(b) Wave cancellation type

In this type the burnt gases are separated and again combined and expelled out to the atmosphere. In this type the speed of the gas can be reduced but the sound can't be reduced completely. Fig 6.12.3(b) shows the wave cancellation type.

6.12.3(c) Resonance type

In this type of mufflers the serially arranged resonators absorb the sound of the out coming gases. So that the sound as well as heat gets reduced. Fig 6.12.3(c) shows the resonance type.

6.12.3(d) Absorber type

In this type the exhaust pipe is surrounded by special type of materials which could absorb the sound. When the exhaust enters into this muffler the absorber material absorbs the sound. Fig 6.12.3(d) shows the absorber type.

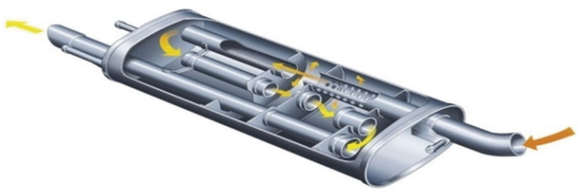


Figure 6.12.3(b) Wave Cancellation Type Muffler

AM_Unit_06.indd 132 31-01-2019 17:49:24